

3 points

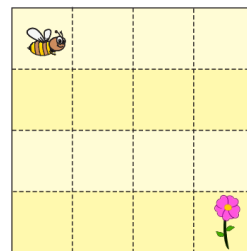
1. Buzz the bee wants to reach the flower.

Which set of directions will get him there.

蜜蜂按照下列哪组方向前进可以采到花蜜?

(A) $\rightarrow \downarrow \rightarrow \downarrow \downarrow \rightarrow$ (B) $\downarrow \downarrow \rightarrow \downarrow \downarrow$ (C) $\rightarrow \downarrow \rightarrow \downarrow \rightarrow$

(D) $\rightarrow \rightarrow \downarrow \downarrow \downarrow$ (E) $\downarrow \rightarrow \rightarrow \downarrow \downarrow \downarrow$

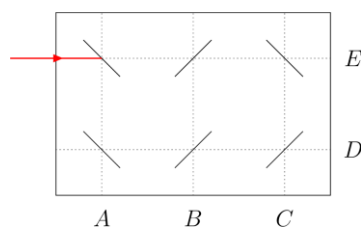
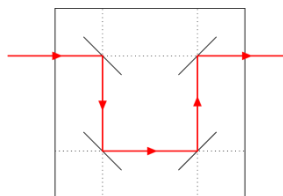


2. Laser beams reflect in mirrors in the way shown in the picture.

At which letter will this laser beam end?

激光束在镜子中的反射情况如右图所示。

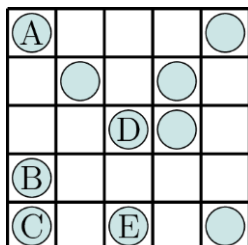
那么下图中的激光束将从哪个字母的位置反射出来?



(A) A (B) B (C) C (D) D (E) E

3. Rossitza wants to put 2 coins in each row and in each column of the grid.

小梦想在网格的每行和每列都放上 2 枚硬币。



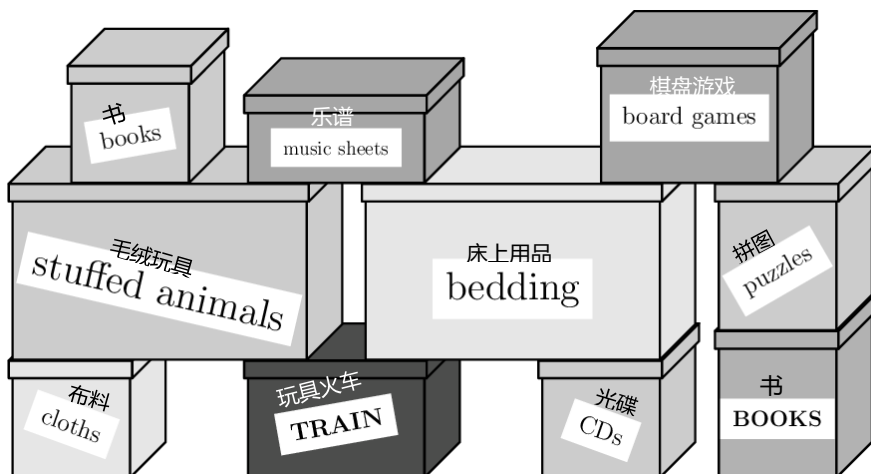
Which coin does she need to move to an empty cell?

请问她现在需要将哪枚硬币移动到其他空格中?

(A) A (B) B (C) C (D) D (E) E

4. What is the smallest number of boxes that Bill has to move to be able to open the dark TRAIN box?

小博至少要移动多少个盒子，才能打开装着玩具火车的黑色盒子？



- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

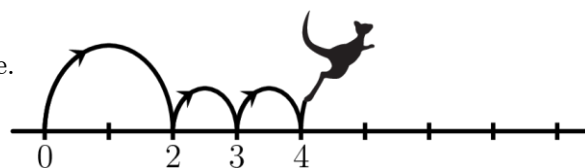
5. Kengu always makes one large jump followed by two small jumps on the number line, as shown in the picture. Kengu starts at 0 and ends on 16.

What is the number of jumps that Kengu makes?

如图所示，袋鼠总是按照先一大跳再两小跳的规律在一条数轴上跳跃。

它从数字 0 开始，一直跳到数字 16。

请问袋鼠一共跳跃了多少次？



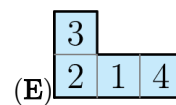
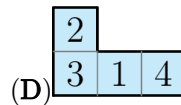
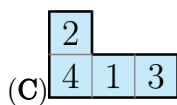
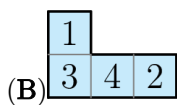
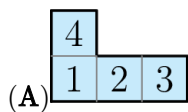
- (A) 4 (B) 7 (C) 8 (D) 9 (E) 12

6. Anna makes a jigsaw where two squares with common sides do not contain the same number. Which piece should she use to complete her jigsaw?

安安制作了一个拼图，其中有公共边的两个小正方形内的数字都不相同。

请问她应该用下面哪一块来完成她的拼图？

3	2	5	4	2	1
1	4	3	1	3	4
2	5		5	2	1
4	1				3
3	2	4	2	5	2
4	1	3	1	3	4



7. $2022 + \square = 2020 + \square$

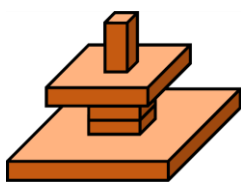
Which two numbers can be written in the two boxes to make the statement correct?

下面哪两个数字可以填入两个方框中使等式成立?

- (A) 3 and 5 3 和 5 (B) 4 and 1 4 和 1 (C) 3 and 4 3 和 4
(D) 7 and 2 7 和 2 (E) 9 and 8 9 和 8

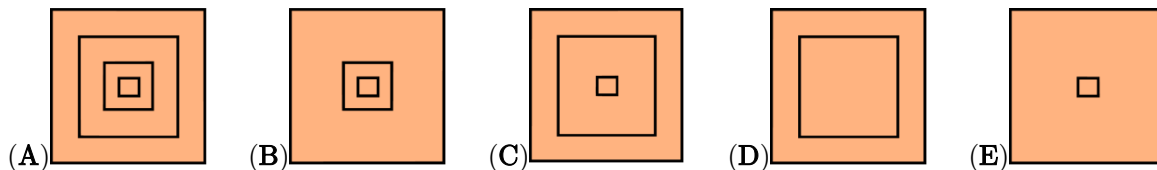
8. John builds the tower shown.

小吉搭了一座塔，如图所示。



What will he see if he looks at his tower from above?

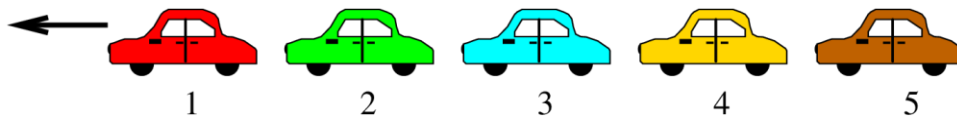
如果从上方看这座塔，可以看到下面哪个图案?



4 points

9. Five cars numbered 1, 2, 3, 4 and 5 are moving in the same direction.

五辆编号为 1、2、3、4、5 的汽车朝同一方向行驶。



First, the last car (5) overtakes the two cars ahead of it.

Next, the second last car overtakes the two cars ahead of it.

Finally, the middle car overtakes the two cars ahead of it.

In what order are the cars now?

首先，最后一辆车 (5) 超过了它前面的两辆车。

接下来，倒数第二辆车超过了它前面的两辆车。

最后，中间的车超过了它前面的两辆车。

请问现在汽车的编号顺序是什么?

- (A) 1, 2, 3, 5, 4 (B) 2, 1, 3, 5, 4 (C) 2, 1, 5, 3, 4 (D) 3, 1, 4, 2, 5 (E) 4, 1, 2, 5, 3

10. The ages of a family of kangaroos are 2, 4, 5, 6, 8 and 10 years.

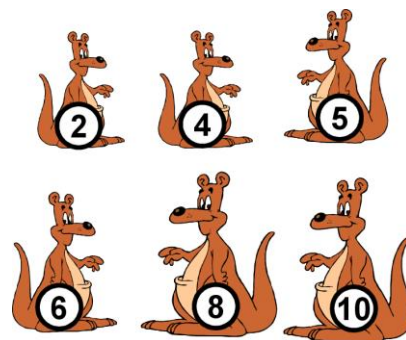
The sum of the ages of four of them is 22 years.

What are the ages of the other two kangaroos?

袋鼠家族中的袋鼠年龄分别是 2、4、5、6、8 和 10 岁。

其中四只袋鼠的年龄之和为 22 岁。

那么另外两只袋鼠的年龄是多少？



- (A) 2 and 8 2 岁和 8 岁 (B) 4 and 5 4 岁和 5 岁 (C) 5 and 8 5 岁和 8 岁
(D) 6 and 8 6 岁和 8 岁 (E) 6 and 10 6 岁和 10 岁

11. During my holiday I sent the five postcards shown below to my friends.

There are **no** ducks on Mike's card.

Cara's card has the sun on it.

There are exactly two living creatures on Paula's card.

Lexi's card has a dog on it.

There are kangaroos on Heather's card.

Which card did Mike get?

假期里，我把下面五张明信片寄给了我的朋友们。

没有鸭子的明信片是小麦的。

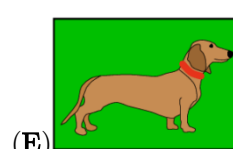
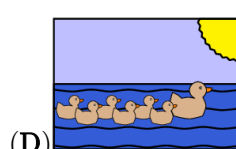
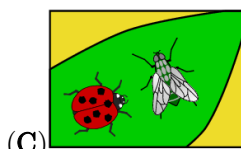
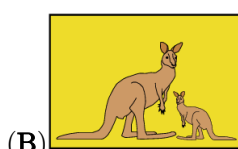
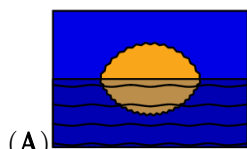
拉拉收到的明信片上有太阳。

只有两个生物的明信片是宝儿的。

小雷收到的明信片上有一只狗。

有袋鼠的明信片是希希的。

请问小麦收到的是下面哪张明信片？



12. Mosif wanted the sum of the three numbers in each row and in each column of the grid to be the same.

He made one mistake.

Which number must he correct?

小莫想让网格中每一行和每一列的三个数字之和都相同。

但他犯了一个错误。

请问他必须修改哪个数字？

9	1	5
3	7	6
4	7	4

- (A) 1 (B) 3 (C) one of the 4s 其中一个数字 4
(D) 5 (E) one of the 7s 其中一个数字 7

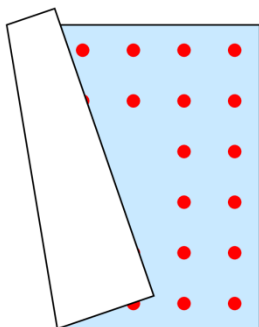
13. Aladdin has a square carpet.

There are the same number of dots, arranged in two lines, along each side of his carpet. Unfortunately, the carpet has folded.

阿拉丁有一块正方形地毯。

地毯每一侧都有两行相同数量的圆点。

但地毯被折叠了。



How many dots are there on Aladdin's carpet?

请问阿拉丁的地毯上一共有多少个圆点？

- (A) 48 (B) 44 (C) 40 (D) 36 (E) 32

14. Joanna folds the number square twice as shown.

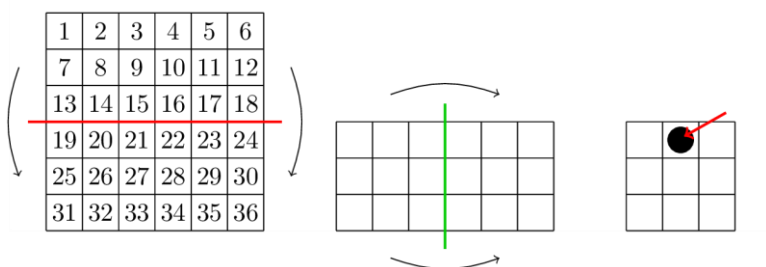
Then she punches a hole through the black spot shown by the arrow.

Which numbers does she also punch through?

小乔将数字方格纸按照如图所示的方式折叠了两次。

然后她在箭头指向的黑点上打了一个洞。

请问这个洞穿过了哪些数字？



- (A) 8, 11, 26, 29 (B) 14, 17, 20, 23 (C) 15, 16, 21, 22
(D) 14, 16, 21, 23 (E) 15, 17, 20, 22

15. The pupils in a class sit in rows.

There are the same number of pupils in each row.

There are 2 rows of pupils in front of Robert and 1 row of pupils behind him.

In his row, there are 3 pupils on his left and 5 pupils on his right.

How many pupils are there in this class?

某班的学生在教室排排坐。

每排的学生人数相同。

小罗前面有 2 排同学，后面有 1 排同学。

在他那一排中，他的左边 3 名同学，右边有 5 名同学。

请问这个班一共有多少名学生？

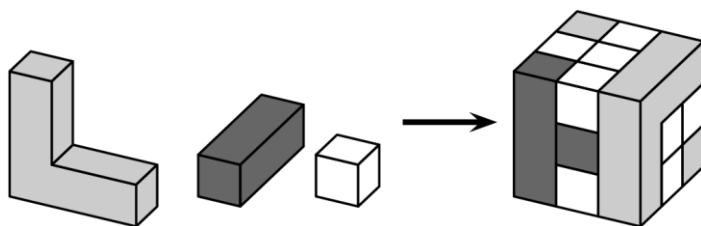
- (A) 10 (B) 17 (C) 18 (D) 27 (E) 36

16. The cube in the picture is built from the three kinds of wooden blocks shown.

How many white wooden blocks are used?

一个立方体由三种木块组成，如图所示。

请问总共用了多少个白色木块？



- (A) 8 (B) 11 (C) 13 (D) 16 (E) 19

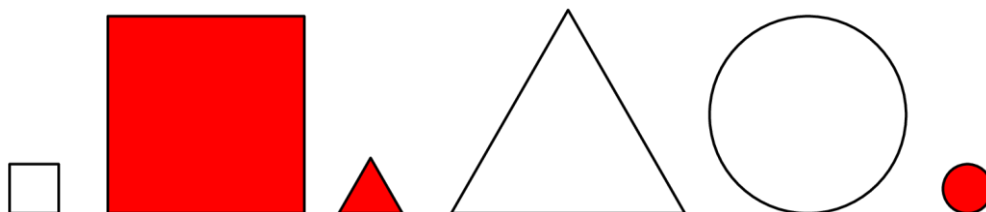
5 points

17. Wanda chose a few of the following shapes and said "Amongst the shapes I have chosen, there are 2 coloured ones, 2 large ones and 2 round ones."

What is the smallest number of the following shapes that Wanda could have chosen?

小达从以下形状中挑选出几个，然后说：“我选出的形状中有 2 个涂色的，2 个大的，2 个圆的。”

请问小达最少可以从以下形状中选出几个？



- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

18. Three football teams participate in a sports tournament.

Each team plays the other two teams exactly once.

In each game, the winner gets 3 points and the loser doesn't get any points.

If the game finishes in a draw, each team gets 1 point.

At the end of the tournament, which number of points is it **impossible** for any team to have?

三支足球队参加体育比赛。

每支球队与另外两支球队只交手一次。

每场比赛中，胜方得 3 分，败方不得分。

如果比赛以平局结束，每队得 1 分。

在比赛结束时，任何球队都**不可能**得到的分数是？

- (A) 1 (B) 2 (C) 4 (D) 5 (E) 6

19. A pyramid is built from cubes with a side-length of 10 cm.

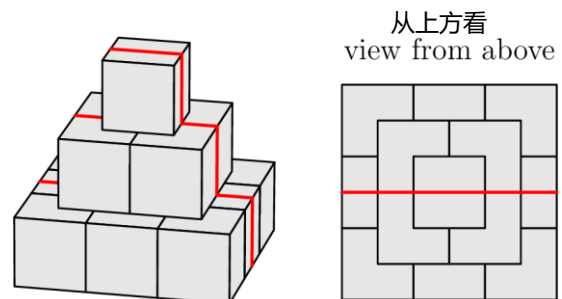
An ant climbed up and over the pyramid, as shown by the red line.

What is the length of the path walked by the ant across the pyramid?

一个金字塔是由边长为 10 厘米的小立方体搭建而成的。

一只蚂蚁爬上并越过金字塔，路径如图中红线所示。

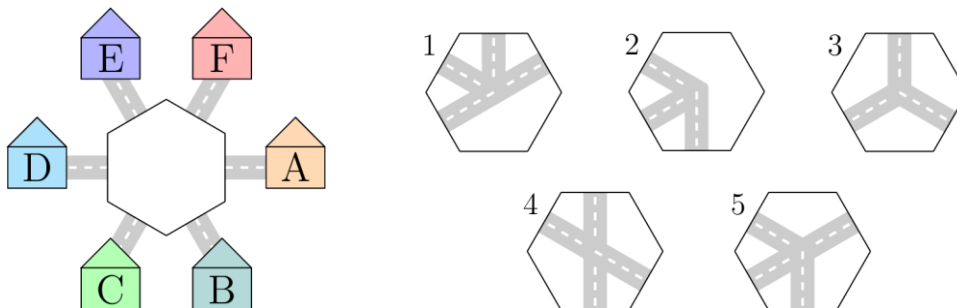
这只蚂蚁经过金字塔的路径长度是多少？



- (A) 30 cm 30 厘米 (B) 60 cm 60 厘米 (C) 70 cm 70 厘米
(D) 80 cm 80 厘米 (E) 90 cm 90 厘米

20. Alma wants to put one of the pieces shown in the middle of the picture so that a child in A is able to travel to B and to E, but not to D. She can rotate the pieces.

小玛想把下图所示的其中一块碎片放在图片中间，这样 A 区的孩子就可以去 B 区和 E 区游玩，但不能去 D 区。碎片可以旋转。

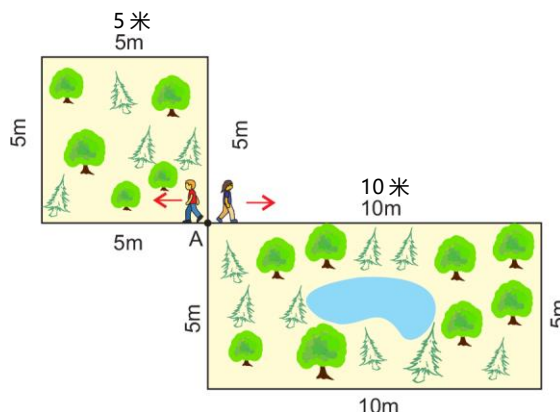


Which two pieces could she use?

请问她可以用哪两块碎片？

- (A) 1 and 2 1 和 2 (B) 2 and 3 2 和 3 (C) 1 and 4 1 和 4
(D) 4 and 5 4 和 5 (E) 1 and 5 1 和 5

21. Ahmad and Zhaleh start moving from point A with the same speed, in the directions shown. Ahmad walks around the square-shaped garden and Zhaleh walks around the rectangular-shaped one. They meet again at A. What is the smallest number of laps around the square-shaped garden that Ahmad could do to meet Zhaleh there?



小艾和小扎从点 A 开始以相同的速度沿图中所示方向前进。
小艾绕着正方形的花园走，小扎绕着长方形的花园走。
他们在点 A 再次相遇。
小艾最少要绕几圈正方形花园才能在点 A 遇见小扎？

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

22. Five children ate some plums.

Lauren ate two plums more than Sophie.
Betty ate three plums fewer than Lauren.
Claire ate one plum more than Betty and three plums fewer than Alice.
Which two girls ate the same number of plums?

五个孩子吃了一些李子。

L 比 S 多吃了两颗李子。

B 比 L 少吃了三颗李子。

C 比 B 多吃一颗李子，比 A 少吃三颗李子。

请问哪两个孩子吃了相同数量的李子？

- (A) Claire and Lauren. C 和 L (B) Claire and Sophie. C 和 S
(C) Lauren and Alice. L 和 A (D) Sophie and Alice. S 和 A
(E) Alice and Betty. A 和 B

23. The small caterpillar shown in the picture curls up to sleep.

图中的毛毛虫蜷缩起来睡觉。



What might that look like?

可能会是什么样子的？

- (A) (B) (C) (D) (E)

24. In the grid, the same number is hidden under the same colour square.

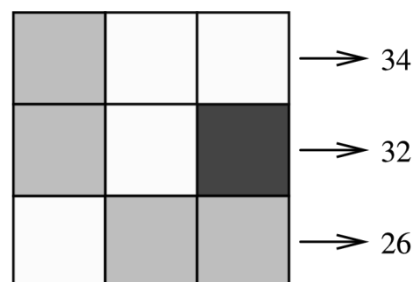
To the right of each row, the sum of the numbers hidden under the squares in that row is given.

Which number is hidden under the black square?

右图网格中，相同颜色的方块下面隐藏有相同的数字。

每一行右边的数字表示隐藏在该行方块下的数字之和。

请问哪个数字隐藏在黑色方块下面？

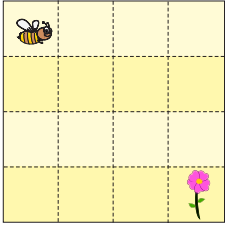


- (A) 6 (B) 8 (C) 10 (D) 12 (E) 14

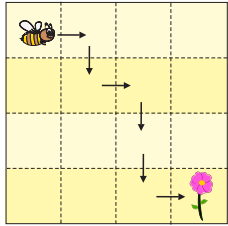
3 points

1 (KW). Buzz the bee wants to reach the flower. Which set of directions will get him there.

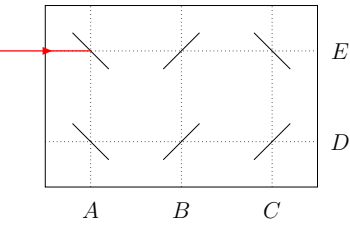
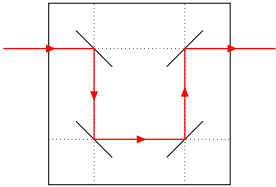
- (A) → ↓ → ↓ ↓ →
- (B) ↓ ↓ → ↓ ↓
- (C) → ↓ → ↓ →
- (D) → → ↓ ↓ ↓
- (E) ↓ → → ↓ ↓ ↓



SOLUTION: The correct answer is (A) - three moves right and three moves down (right, down, right, down, down, right):

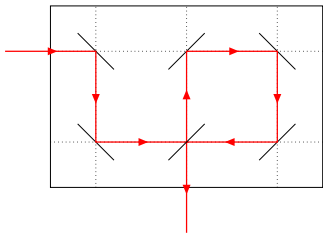


2 (DK). Laser beams reflect in mirrors in the way shown in the picture. At which letter will this laser beam end?

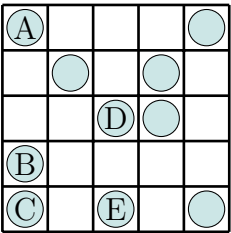


- (A) A
- (B) B
- (C) C
- (D) D
- (E) E

SOLUTION: The laser beam travels down, then right, up, right, down, left, down and ends up at point B.



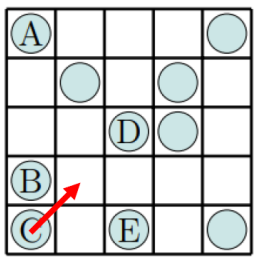
3 (DK). Rossitza wants to put 2 coins in each row and in each column of the grid.



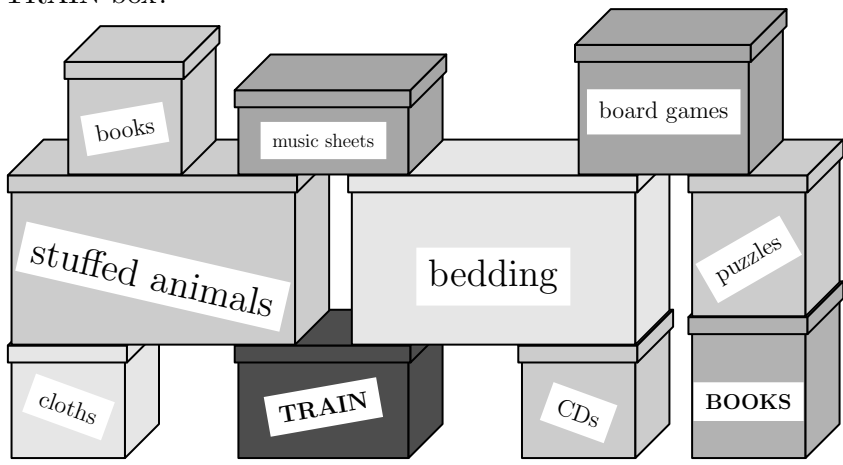
Which coin does she need to move to an empty cell?

- (A) A (B) B (C) C (D) D (E) E

SOLUTION: Notice that most rows and columns have two coins, while the first column and the last row have 3 coins each for which the common coin is the coin C. This coin needs to be moved to the column/row with only one coin which is the second column fourth row (moving coin C diagonally next to the coin B:



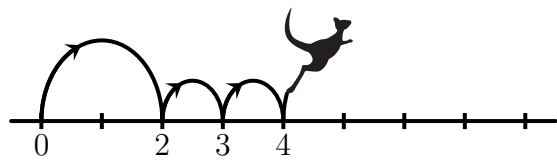
4 (DE). What is the smallest number of boxes that Bill has to move to be able to open the dark TRAIN box?



- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

SOLUTION: In order to reach the train box, Bill must move the boxes with books, music sheets, board games, stuffed animals and bedding - a total of 5 boxes.
 OR: He has to move 1. Books, 2. Music Sheets, 3. Board Games, 4. Stuffed Animals and 5. Bedding. The total is 5.
 Note: Only dark OR train is needed. We leave both for now for translation reasons.

5 (NO). Kengu always makes one large jump followed by two small jumps on the number line, as shown in the picture.
 Kengu starts at 0 and ends on 16.
 What is the number of jumps that Kengu makes?



- (A) 4 (B) 7 (C) 8 (D) 9 (E) 12

SOLUTION: The sum of the distance of one large jump and two small jumps is 4. It is repeated. Therefore $16 \div 4 = 4$ and $4 \times 3 = 12$.

3	2	5	4	2	1
1	4	3	1	3	4
2	5		5	2	1
4	1				3
3	2	4	2	5	2
4	1	3	1	3	4

6 (DK).

Anna makes a jigsaw where two squares with common sides do not contain the same number. Which piece should she use to complete her jigsaw?

- (A)

4
1 2 3
- (B)

1
3 4 2
- (C)

2
4 1 3
- (D)

2
3 1 4
- (E)

3
2 1 4

SOLUTION: The correct answer is D. To persuade yourself in this it is enough to check that each square from the inserted piece D is not neighbor to the square with the same number. On the other hand, all other answers are wrong. If we put the piece A the square with number 1 would have a neighbor on its left with the same number 1. For other pieces the squares with the same neighbor are : square with number 2 for piece B, square with number 4 for piece C and square with number 3 for piece E.

3	2	5	4	2	1
1	4	3	1	3	4
2	5		5	2	1
4	1				3
3	2	4	2	5	2
4	1	3	1	3	4

Solution 2: The neighboring tiles of the red circle are 2, 3 and 5. So the red circle have to be 1 or 4. There is no 1 in the given answers. So it is 4. There are two 4s (D) and (E). The neighboring tiles of green circle are 3 and 5. So the green circle have to be 1, 2 and 4. It is 3 in answer (E) and 2 in answer (D).

7 (CH). $2022 + \square = 2020 + \square$

Which two numbers can be written in the two boxes to make the statement correct?

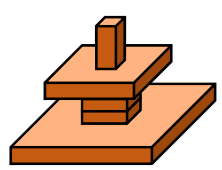
- (A) 3 and 5
- (B) 4 and 1
- (C) 3 and 4
- (D) 7 and 2
- (E) 9 and 8

SOLUTION: Notice that the number 2022 is larger than 2020 by 2, hence in order for the equation to be correct the number in the first square should be smaller than the number in the second by 2. Among the offered pairs of numbers, the only two numbers that can be put in squares are 3 and 5.

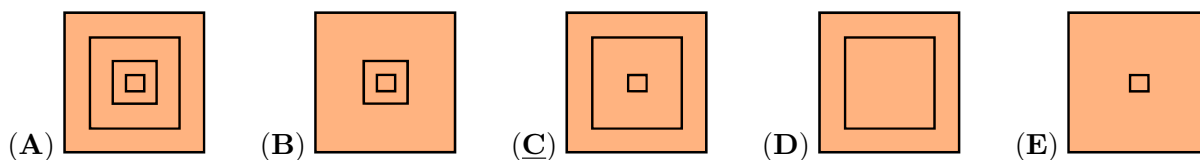
OR: $2022 + 3 = 2020 + 5 (= 2025)$

Note: if a country thinks this question is not suitable for 3p as numbers ; 1000 are needed, they can change the question into $20 + (\text{box}) = 22 + (\text{box})$ and swap all answers (so the first answer is then 5 and 3, and so on). Another option is to move this question to the 4p or 5p section as for 4p or 5p the year can be an exception.

8 (GR). John builds the tower shown.



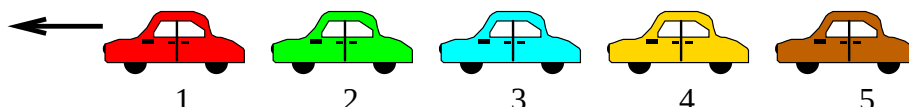
What will he see if he looks at his tower from above?



SOLUTION: In the picture, five solid figures of different sizes are stacked. When looking down at these solid figures from above, the third and fourth figures are invisible because they are smaller than the second figure.

4 points

9 (PL). Five cars numbered 1, 2, 3, 4 and 5 are moving in the same direction.



First, the last car (5) overtakes the two cars ahead of it.
Next, the second last car overtakes the two cars ahead of it.
Finally, the middle car overtakes the two cars ahead of it.
In what order are the cars now?

- (A) 1, 2, 3, 5, 4 (B) 2, 1, 3, 5, 4 (C) 2, 1, 5, 3, 4 (D) 3, 1, 4, 2, 5 (E) 4, 1, 2, 5, 3

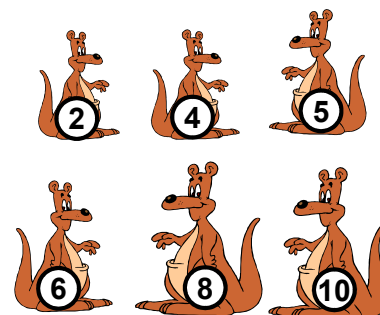
SOLUTION: The order of the cars changes like this: 1, 2, 3, 4, 5 $\rightarrow$ 1, 2, 5, 3, 4 $\rightarrow$ 1, 3, 2, 5, 4 $\rightarrow$ 2, 1, 3, 5, 4

Note: second last (=penultimate)

10 (GR). The ages of a family of kangaroos are 2, 4, 5, 6, 8 and 10 years.

The sum of the ages of four of them is 22 years.
What are the ages of the other two kangaroos?

- (A) 2 and 8 (B) 4 and 5
(C) 5 and 8 (D) 6 and 8
(E) 6 and 10



SOLUTION: The sum of all the ages is $2+4+5+6+8+10 = 35$ so the other two add up to $35 - 22 = 13$. This is uniquely 5 + 8 (easy to spot because only one of the given numbers is odd).

11 (DE). During my holiday I sent the five postcards shown below to my friends.

There are **no** ducks on Mike's card.

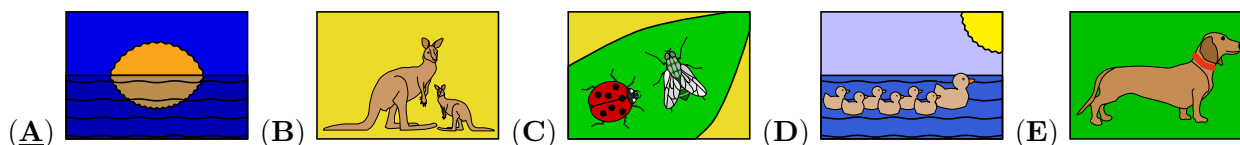
Cara's card has the sun on it.

There are exactly two living creatures on Paula's card.

Lexi's card has a dog on it.

There are kangaroos on Heather's card.

Which card did Mike get?



SOLUTION: The deduction is: Mike is not D, Heather is B, then Paula is C, then Cara is A or D, Lexie is E. Since now B, C, E and D are all excluded for Mike, Mike can only be A (and Cara will be

D).

Remark: There is also a black-and-white version of the pictures.

12 (GR). Mosif wanted the sum of the three numbers in each row and in each column of the grid to be the same.

He made one mistake.

Which number must he correct?

9	1	5
3	7	6
4	7	4

(A) 1

(B) 3

(C) one of the 4s

(D) 5

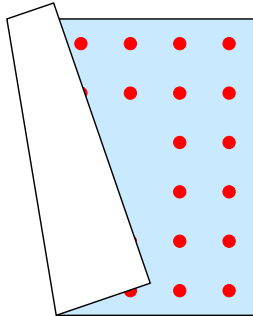
(E) one of the 7s

SOLUTION: The sums in most rows and columns is 15. The exceptions are the first column and the second row where the sum is 16. So we must change the number at their intersection by 1 less.

9	1	5	→ 15
2 3	7	6	→ 16 15
4	7	4	→ 15
↓	↓	↓	
16 15	15	15	

13 (PL). Aladdin has a square carpet.

There are the same number of dots, arranged in two lines, along each side of his carpet. Unfortunately, the carpet has folded.



How many dots are there on Aladdin's carpet?

(A) 48

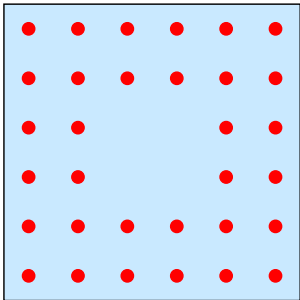
(B) 44

(C) 40

(D) 36

(E) 32

SOLUTION: The same number of dots arranged in two rows means there are two rows of six dots along each side, therefore the unfolded square carpet should be as follows:



The total dots should be $4 \times 6 + 8 = 32$.

Solution 2.

If the whole carpet was covered with dots, it would be $6 \cdot 6 = 36$. Two rows of dots (4 dots) are missing in the middle. In total $36 - 4 = 32$ dots.

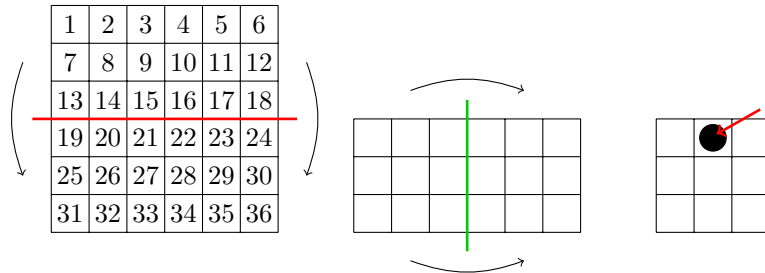
Solution 3.

If the whole carpet was covered with dots, it would be $6 \cdot 6 = 36$. Thus, the only possible answer is E.

14 (GR). Joanna folds the number square twice as shown.

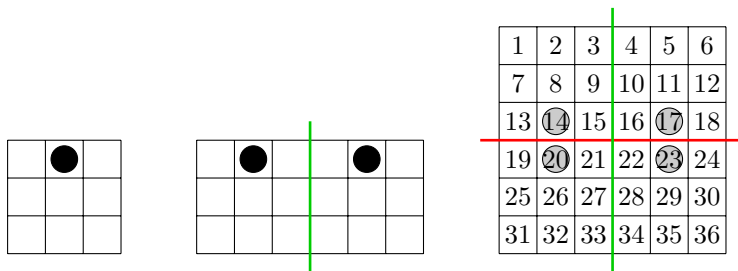
Then she punches a hole through the black spot shown by the arrow.

Which numbers does she also punch through?



- (A) 8, 11, 26, 29 (B) 14, 17, 20, 23 (C) 15, 16, 21, 22 (D) 14, 16, 21, 23 (E) 15, 17, 20, 22

SOLUTION: The picture shows the paper and the holes as we unfold it to its original shape. The numbers punched are clear.



Note: If a country decides to use a dotted/dashed line, please use the official origami notation: see Yoshizawa-Randlett system. Here that would mean: dashed lines (no dots).

15 (PH). The pupils in a class sit in rows.

There are the same number of pupils in each row.

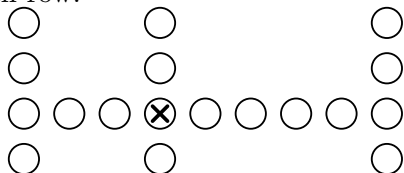
There are 2 rows of pupils in front of Robert and 1 row of pupils behind him.

In his row, there are 3 pupils on his left and 5 pupils on his right.

How many pupils are there in this class?

- (A) 10 (B) 17 (C) 18 (D) 27 (E) 36

SOLUTION: As shown in the picture below, this rectangle formation has $3 + 1 + 5 = 9$ members in each row.

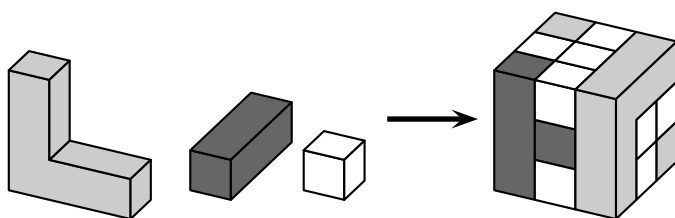


It follows the number of rows in the formation is $2 + 1 + 1 = 4$ rows. Therefore, the total number of pupils is $4 \times 9 = 36$ pupils in all.

Note: As some regard this problem similar to the car problem; it can be replaced with a problem from reserved.

16 (PH). The cube in the picture is built from the three kinds of wooden blocks shown.

How many white wooden blocks are used?



(A) 8

(B) 11

(C) 13

(D) 16

(E) 19

SOLUTION: The first layer has a total of 2 white blocks , while the second layer has a total of 6 white blocks and the third layer has a total of 3 white blocks . Therefore, the total number of white wooden blocks in a cube is $2 + 6 + 3 = 11$.

OR A large cube can be made with a total of 27 small white wooden blocks (cubes). Grey wooden blocks can be formed by 5 white wooden blocks, and black wooden blocks can be formed by 3 white wooden blocks.

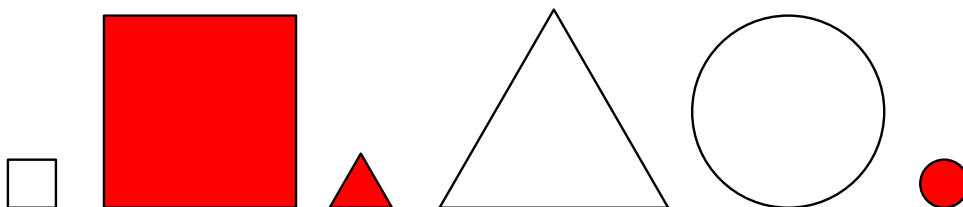
If you look at the picture, you can see that 2 grey wooden blocks and 2 black wooden blocks are used to make a large cube.

Assuming a large cube with 27 small white wooden blocks in total, $27 - 10$ (2 grey blocks) $- 6$ (2 black blocks) $= 11$, so 11 white wooden blocks were used.

5 points

17 (PL). Wanda chose a few of the following shapes and said "Amongst the shapes I have chosen, there are 2 coloured ones, 2 large ones and 2 round ones.

What is the smallest number of the following shapes that Wanda could have chosen?



(A) 2

(B) 3

(C) 4

(D) 5

(E) 6

SOLUTION: Wanda selects the two round ones (circles); one is coloured and one is white. Then she has already one large figure. She could chose the other large coloured piece (square), and all conditions are satisfied. So the minimum amount of figures that Wanda selected is three.

18 (IR). Three football teams participate in a sports tournament.

Each team plays the other two teams exactly once.

In each game, the winner gets 3 points and the loser doesn't get any points.

If the game finishes in a draw, each team gets 1 point.

At the end of the tournament, which number of points is it **impossible** for any team to have?

(A) 1

(B) 2

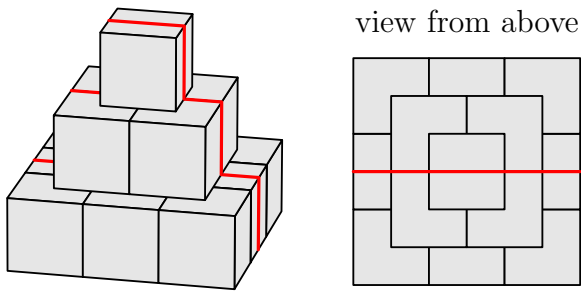
(C) 4

(D) 5

(E) 6

SOLUTION: As every team has just 2 games and the possibility to get 0, 1 or 3 points each game, the only possibilities are $0 + 0 = 0$ points, $0 + 1 = 1$ point, $1 + 1 = 2$ points, $1 + 3 = 4$ points and $3 + 3 = 6$ points. There is no chance to get 5 points - for this case a team would have to have at least 3 games ($3 + 1 + 1$ points).

19 (SK). A pyramid is built from cubes with a side-length of 10 cm.
 An ant climbed up and over the pyramid, as shown by the red line.
 What is the length of the path walked by the ant across the pyramid?



- (A) 30 cm (B) 60 cm (C) 70 cm
 (D) 80 cm (E) 90 cm

SOLUTION: To the middle of the cube in the top of the pyramid the ant has to climb 3 cubes vertically ($3 \cdot 10\text{cm} = 30\text{cm}$) and 3 times half a cube length horizontally ($3 \cdot 5\text{cm} = 15\text{cm}$). The way from the top has the same length since the pyramid is symmetric. So in total the ant will climb $30\text{cm} + 15\text{cm} + 30\text{cm} + 15\text{cm} = 90\text{cm}$.

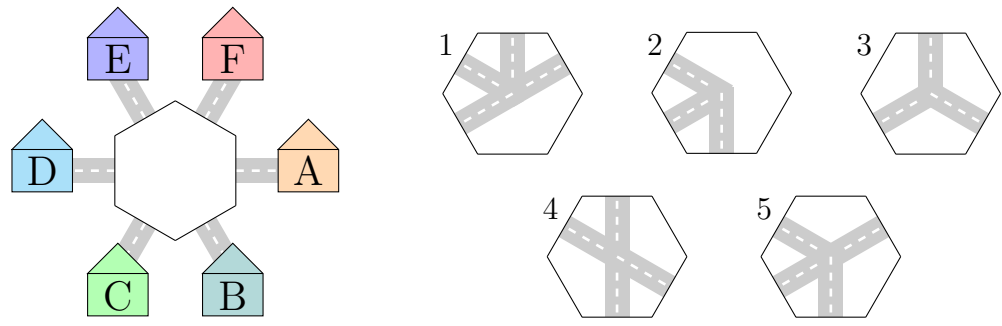
Alternative solution (if we assume that the cubes are arranged symmetrically): After climbing up the first cube (10cm) the ant has to walk horizontally along the red line for the half length of a cube (5cm). So it can easily be counted: 10 (up) + 5 (horizontally) + 10 + 5 + 10 + 10 (straight, on the top) + 10 (down) + 5 + 10 + 5 + 10 = 90 cm.

Solution 3.

The ant's way consists of six vertical intervals (height of a cube) and is $6 \cdot 10 = 60\text{ cm}$. In the view from above we can see the horizontal part of the way of the ant, which in total is equal to three of the cube's edge, i.e. $3 \cdot 10 = 30\text{ cm}$. In total $60 + 30 = 90\text{ cm}$.

Note: If you print in black/white you need to change the word red line. Picture G15331-3 is without the 'view from above' text so that people can choose that for their own language (for translation reasons).

20 (DK). Alma wants to put one of the pieces shown in the middle of the picture so that a child in A is able to travel to B and to E, but not to D.
 She can rotate the pieces.

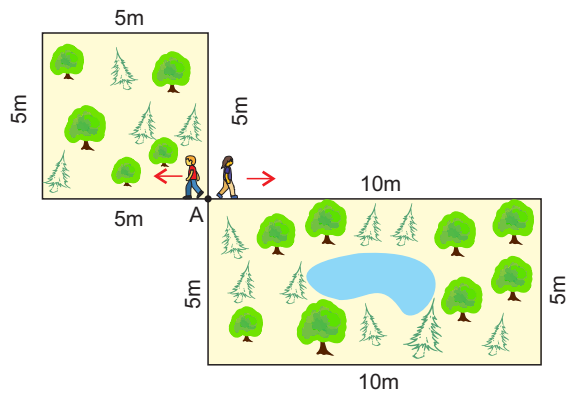


Which two pieces could she use?

- (A) 1 and 2 (B) 2 and 3 (C) 1 and 4 (D) 4 and 5 (E) 1 and 5

SOLUTION: 1 can be put in so that the straight street leads from B to E, then A (and F, what doesn't matter) has a connection as well. 2 doesn't work because this piece connects three neighbours and A, B and E aren't three neighbours. 3 would connect A with C and E, so not to B. 4 doesn't work because as soon as A has access, it has access straight to D. 5 works if you put the "lonely" end of the streets to E, so A, B and C are connected as well, but not D. So we note that 1 and 5 are the only two pieces which can be put in the space so that A can travel to B and E, but not to D.

21 (IR). Ahmad and Zhaleh start moving from point **A** with the same speed, in the directions shown. Ahmad walks around the square-shaped garden and Zhaleh walks around the rectangular-shaped one. They meet again at **A**. What is the smallest number of laps around the square-shaped garden that Ahmad could do to meet Zhaleh there?



- (A) 1 (B) 2 (C) 3
(D) 4 (E) 5

SOLUTION: When Ahmad has returned to point *A* for the first time, he has walked $4 \cdot 5\text{m} = 20\text{m}$, but then Zhaleh is not at point *A*, as he must walk $2 \cdot 5\text{m} + 2 \cdot 10\text{m} = 30\text{m}$ for a complete round. When Ahmad has returned to point *A* for the second time, he has walked $2 \cdot 20\text{m} = 40\text{m}$ and Zhaleh has walked more than one but less than 2 rounds, so he is not at point *A*. When Ahmad has returned to point *A* for the third time, he has walked $3 \cdot 20\text{m} = 60\text{m}$. Since $60\text{m} = 2 \cdot 30\text{m}$, in this moment Zhaleh has completed 2 full rounds and is back at point *A*.

Note: We kept the text as short as possible. To be precise, you may want to add: "for the first time after the start" in the last sentence. (Since they meet at the start first, the correct answer would be 0). However, there is no 0 laps as an option right now and this clarification is not necessary if you keep the distractors as proposed (1, 2, 3, 4, 5). therefore, we did not add this sentence now.

22 (PL). Five children ate some plums.
Lauren ate two plums more than Sophie.
Betty ate three plums fewer than Lauren.
Claire ate one plum more than Betty and three plums fewer than Alice.
Which two girls ate the same number of plums?

- (A) Claire and Lauren. (B) Claire and Sophie. (C) Lauren and Alice.
(D) Sophie and Alice. (E) Alice and Betty.

SOLUTION: **Note:** To solve the problem it may be helpful to mark the numbers of plums on a number line.

Lauren ate 2 plums more than Sophie.

Because Betty ate 3 plums less than Lauren, Betty had 1 plum less than Sophie.

Since Claire ate 1 plum more than Betty, Claire ate as many plums as Sophie, so (B) is correct.

(As Alice ate 3 plums more than Claire, Alice ate 1 plum more than Lauren. And Betty and Claire did not eat the same number of plums, So all the numbers are different except for Claire and Sophie.)

23 (DE). The small caterpillar shown in the picture curls up to sleep.



What might that look like?

- (A) (B) (C) (D) (E)

SOLUTION: As you see, the caterpillar has alternately one yellow and one black circle. So you have to find a possible "path" through the sleeping caterpillar where you can draw one line from the head to a black circle, then to a yellow, then to a black and so on. The only possibility to find such a way

through whole the body of the caterpillar is in option A as shown: or

24 (PL). In the grid, the same number is hidden under the same colour square.

To the right of each row, the sum of the numbers hidden under the squares in that row is given.

Which number is hidden under the black square?

- (A) 6 (B) 8 (C) 10
 (D) 12 (E) 14

			→ 34
			→ 32
			→ 26

SOLUTION: The number hidden by grey square plus two numbers hidden by white squares add up 34, according to the first line of the table. The number hidden by white square plus two numbers hidden by grey squares add up 26, according to the third line of the table. This means that three numbers under grey squares plus three numbers under white squares add up $34 + 26 = 60$ (first and third lines). Thus, the number under white square plus the number under grey square equals $60 : 3 = 20$. Finally, if the numbers in the second line sums 32, the number under black must be $32 - 20 = 12$.